CYCLISTIC CASE STUDY: HOW TO INCREASE BIKE ANNUAL MEMBERS.

October 29, 2026

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- **BUSINESS GOAL:** Increase the annual members

  This is because Cyclistic’s finance concluded that annual members are much profitable than casual riders.

- **BUSINESS TASK:**

  Determine how annual members and casual riders use Cyclistic bikes differently, in order to inform marketing strategies to convert casual riders to annual members.

ABOUT THE DATA

- The data has been made available by Motivate International Inc under a data License Agreement.
- Was conducted on 1st party basis indicating high integrity
- Consists of Divvy 2019 Q1 and Divvy 2020 Q1 datasets which consits of 790k+ rows.
- There are two types of riders: Member (Annual) and Casual (single or day pass).

PROCESS:DATA CLEANING

- Tools Used: Python (Pandas) for combining, cleaning and analysis and Tableau for visualization and a final Dashboard.
- Dropped columns (trip_id, bikeid) and (ride_id,rideable_type, lat and lng) in the 2019 Q1 and 2020 Q1 datasets respectively in order to have consistent columns that will be useful in analysis.
- Made columns consistent by renaming usertype to member_casual
- Standardized values: “Subscriber” replaced with “Member” and “Customer” with “Casual”, with counts, and check that only two values remain.

PROCESS:DATA CLEANING

- Combined the two datasets after making the columns consistent.
- Creating new Columns: day_of_week and hour.
- Fixing the ride_length since the exported column contained ### values , so ride length was rebuilt from “started_at” and “ended_at”
- Removed invalid records: 25 rows with negative ride_length
- Changed Datatypes for the member_casual column to categorical and those that included time to datetime datatype.
- Performed verification checks such as value_counts() showing only two rider types, a minimum ride length of zero and row counts that add up.

ANALYSIS 1/5

- I compared annual members and casual riders across 790k+ cleaned rides.
- Using Python, i grouped rides by rider type and measured ride length, day of week, hour of day, start station and round trips.
- Results were visualized in Tableau

- Ride Length

  The median casual ride lasts 22 minutes, compared with 8 min for members, so the typical casual ride is nearly 3x longer. I used median instead of the mean so a few long rides wouldn’t distort the comparison

ANALYSIS 2/5

- Day of week

  Members take most rides on every day, with the highest volumes on weekdays. Casual riders take the larger share of rides on the weekend 21.2 compared to 6.2 during the weekdays.

- Time of day

  Member rides peak at commute hours (7-8am and 4-5pm), while casual rides build gradually and peak in the afternoon (12pm)

ANALYSIS 3/5

- Stations

  The two groups rides at largely different stations, and only Michigan Ave & Washington St appears in both top 10 lists. Casual riders top stations include (HQ QR ,Streeter Dr & Grand Ave, Shedd Aquarium and Millennium Park); members' are concentrated in downtown including(Canal St & Adams St and Clinton St & Washington Blvd)

- Round Trips

  18.8% of casual riders start and end at the same station, versus 1.9% for members, about 10x higher

ANALYSIS 4/5

Conclusion:

1. Members use the bikes in a way consistent with commuting: short, point-to-point trips at commute hours on weekdays.

2. Casual riders' pattern is consistent with leisure: longer rides, more weekend use, afternoon peaks, loop trips and visitor-area stations.



ANALYSIS 5/5

Limitations:

1. The analysis covers only 2019 and 2020 Q1, so seasonal conclusions can’t be drawn.

2. The data has no demographics, pricing or bike type, so it shows how riders behave, not why.

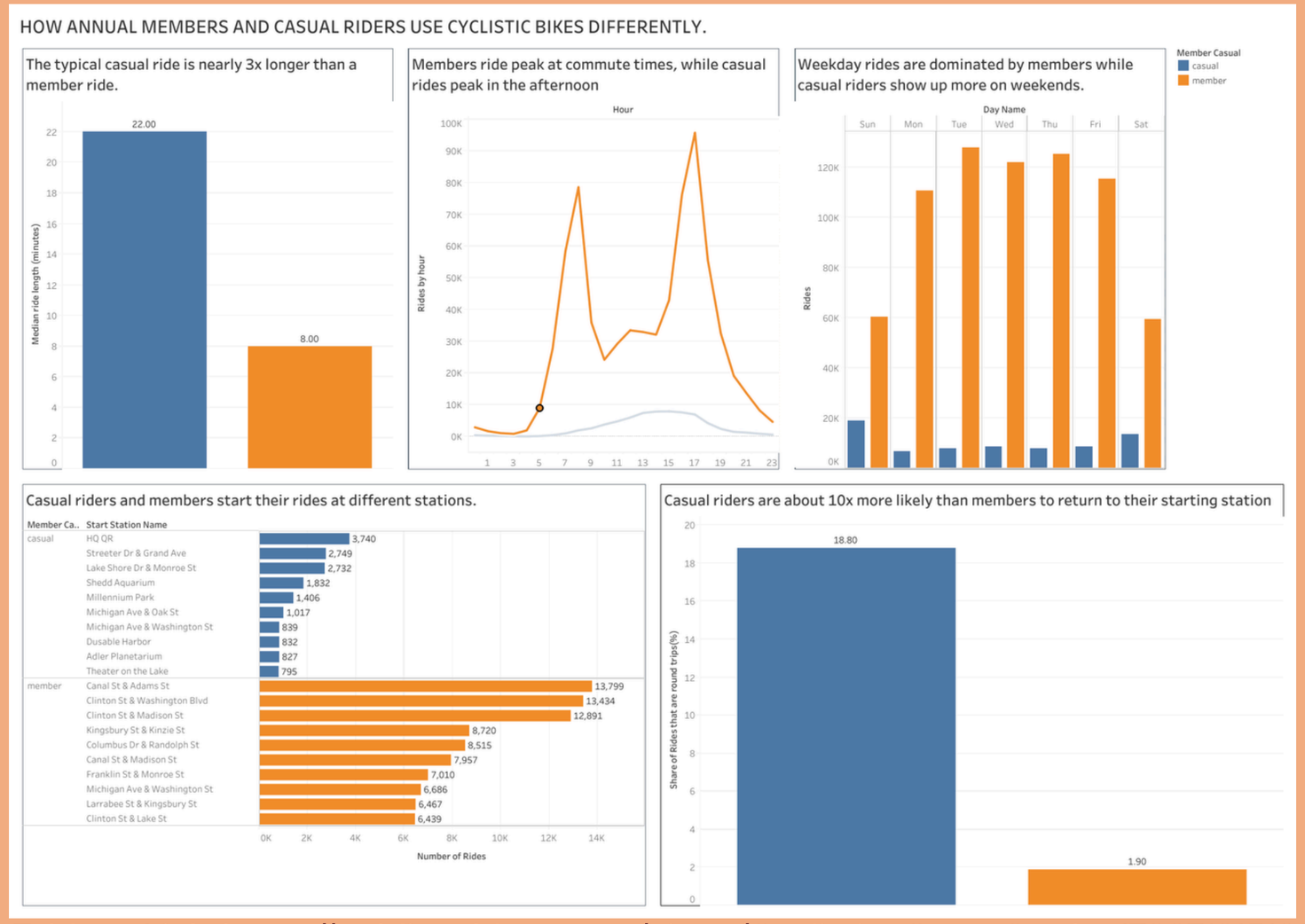
Analysis done on python can be found in the link below

https://github.com/yabasahilda2004/Cyclistic_case_study/blob/main/cyclistic.ipynb



DASHBOARD OVERVIEW:

- How Members and Casual Riders use Cyclistic Bikes Differently.



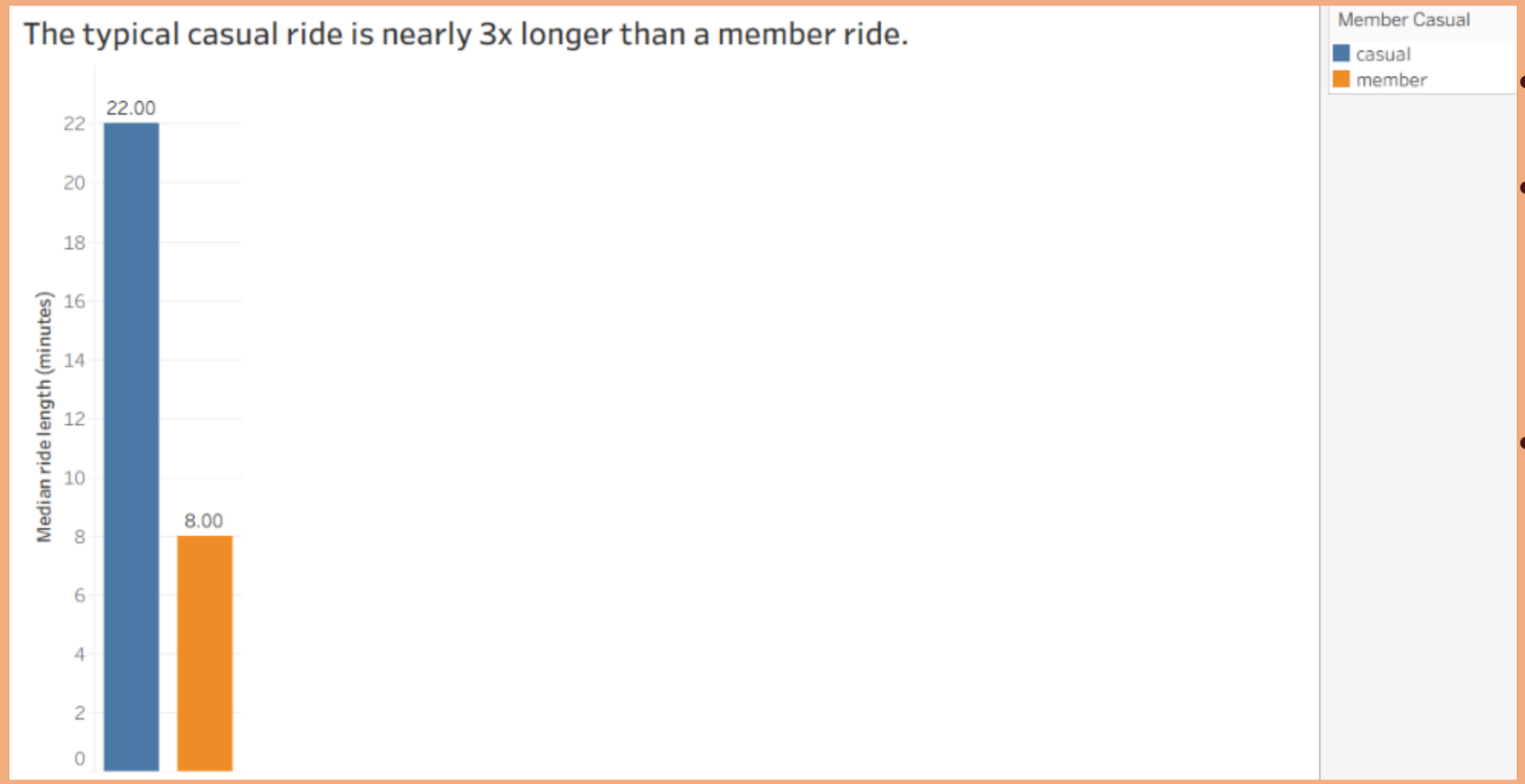
- Interactive dashboard comparing annual members and casual riders by ride length, day of week, time of day, start stations and round trips.

- Source: Cyclistic trip data (Divvy, Motivate International Inc), Q1 2018 and Q1 2020, 790k+ rides after cleaning.

- Tableau Link: https://public.tableau.com/views/Cyclistic_Case_Study_17937121519820/Dashboard?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

CASUAL RIDERS TYPICAL RIDE IS NEARLY 3X LONGER THAN A MEMBERS

Bar Chart of median ride length by rider type.

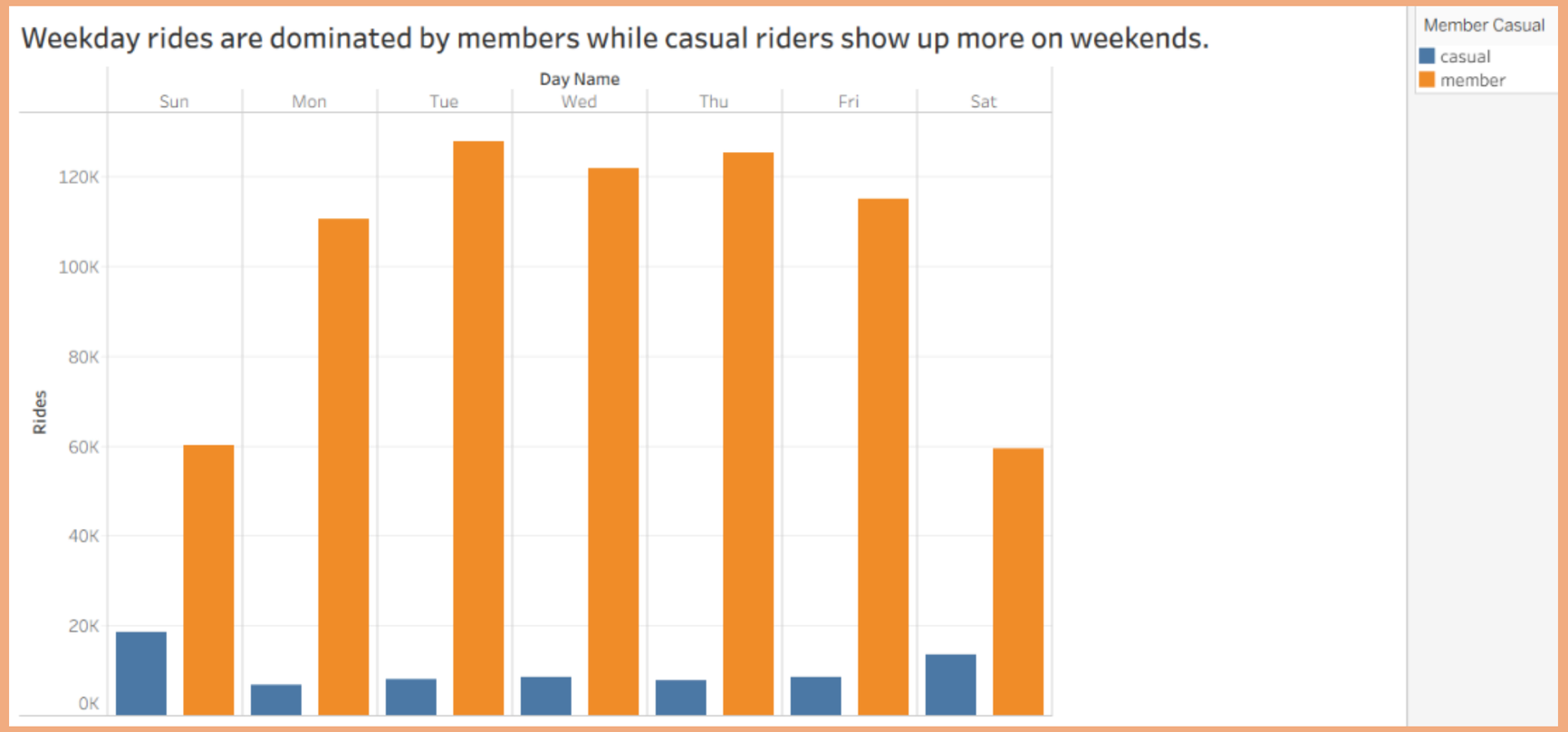


- The median casual ride lasts 22 minutes, compared with 8 minutes for members.
- The median is used so a few very long rides don’t distort the comparison.

- Longer rides are consistent with leisure use, while short rides fit quick, point-to-point trips. This suggests that casual riders value the bikes for recreation.

MEMBERS DOMINATE WEEKDAY RIDES, WHILE CASUAL RIDERS SHOW UP MORE ON WEEKENDS

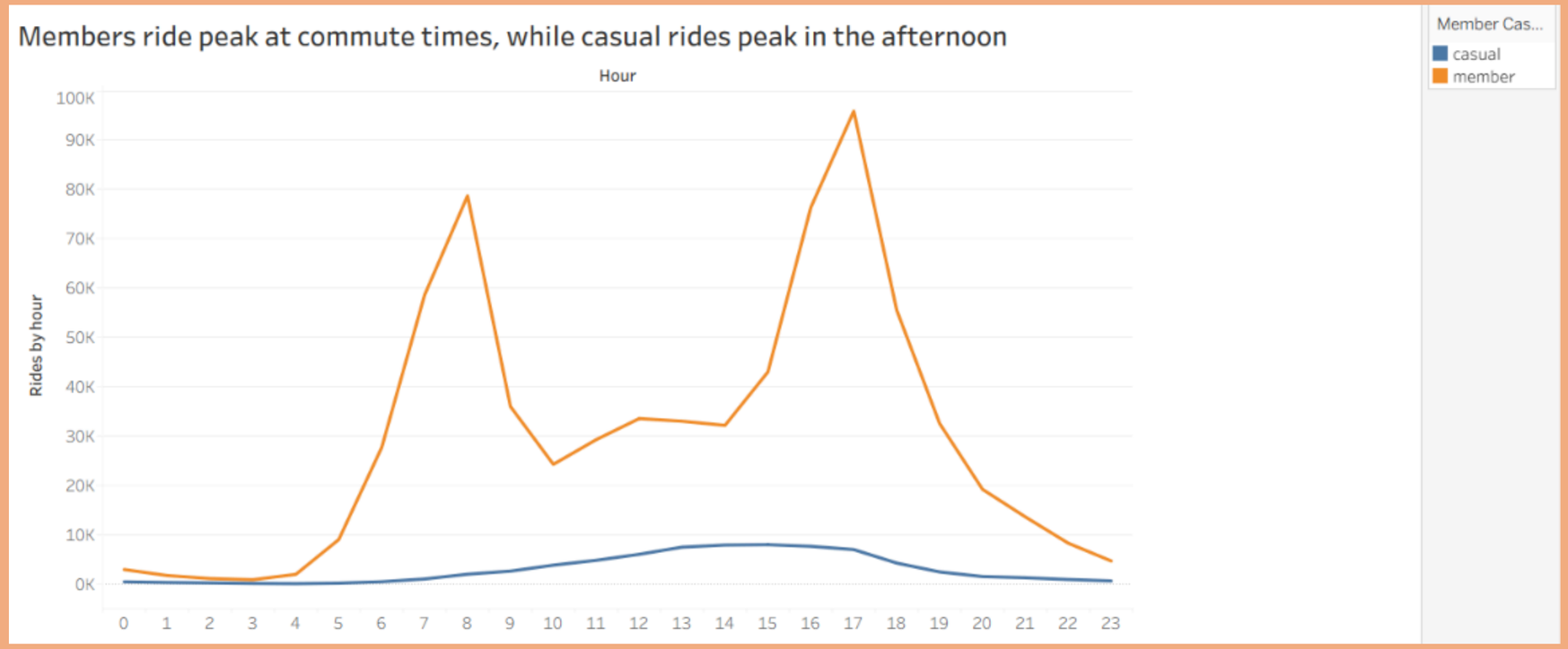
Bar Chart of rides by day of week for casual riders and members.



- Members take far more rides than casual riders on every day, with the highest volumes on weekdays.
- Casual riders make up 21.2% of weekend rides versus 6.2% of weekday rides.
- The weekend share is consistent with leisure riding, while members' weekday volume is consistent with commuting.

MEMBERS RIDE AT COMMUTE TIMES, WHILE CASUAL RIDERS PEAK IN THE AFTERNOON

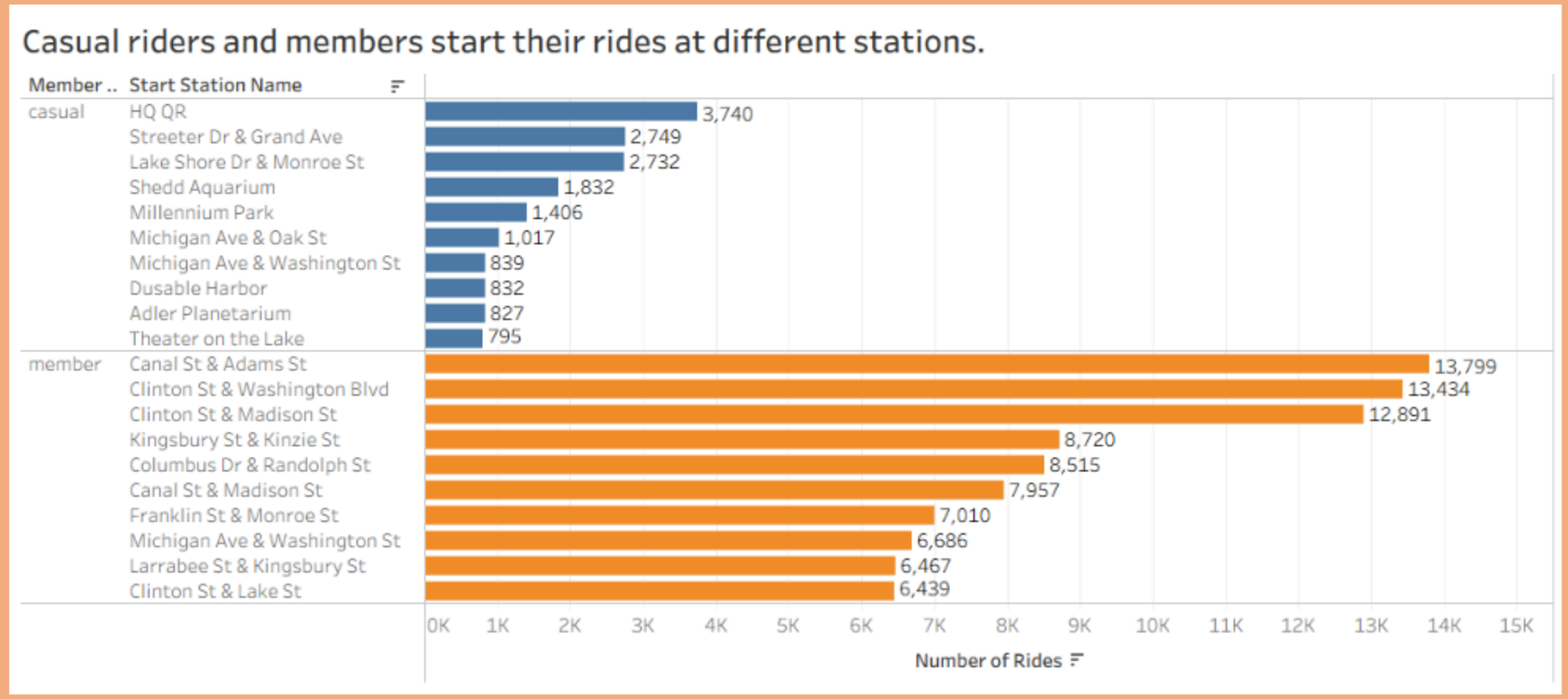
Line Chart of rides by hour of day.



- Member rides peak twice, around 8am and 5pm, which is consistent with commuting.
- Casual rides build up gradually through the day and peak around 1pm.
- Members two daily peaks fit a commute pattern, while casual ride afternoon peak fits leisure or sightseeing use.

CASUAL RIDERS AND MEMBERS START THEIR RIDES AT DIFFERENT STATIONS

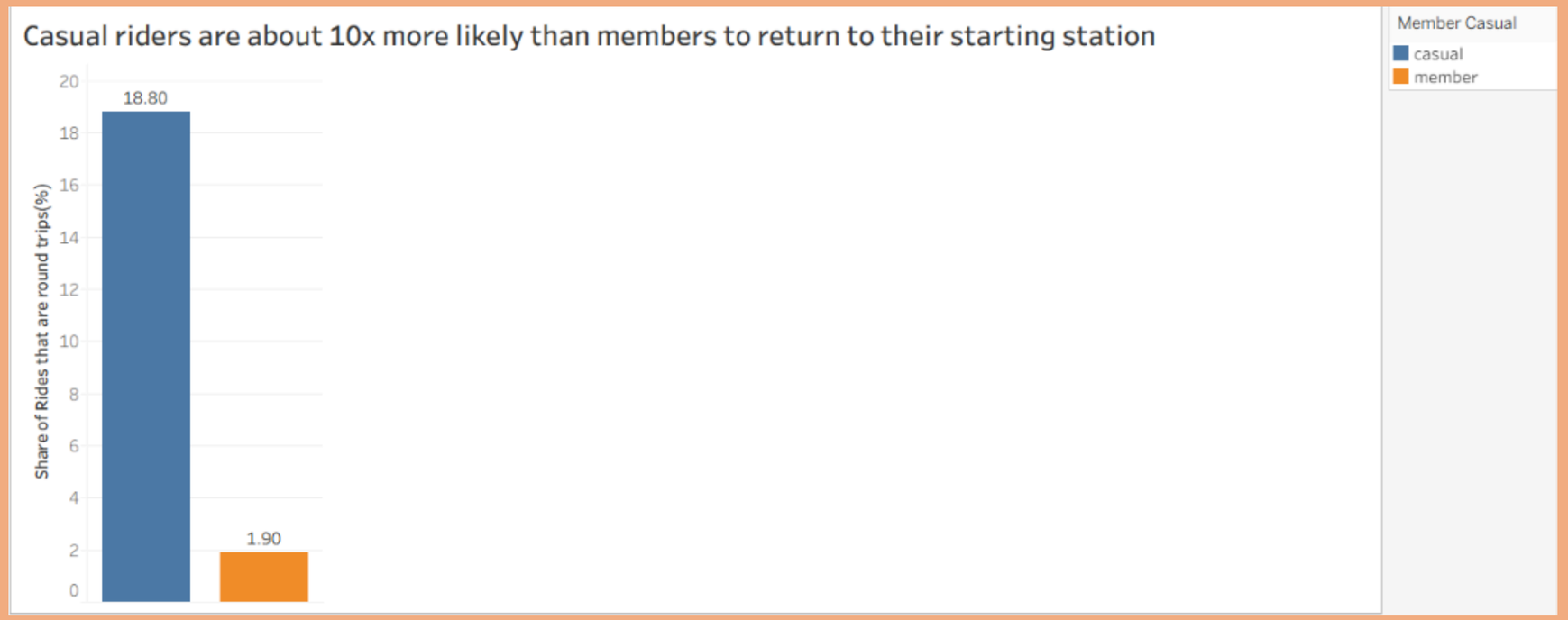
Horizontal bar chart of the top 10 start stations for casual riders and members.



- Only one station, Michigan Ave & Washington St, appears in both top-10 lists.
- Casual riders top stations such as Lake Shore and Shedd Aquarium are popular visitor location.
- Members top stations such as Canal St and Clinton St are concentrated downtown.
- Casual riders stations are consistent with leisure and visitor use, while members are consistent with commuting.
- Members take far more rides overall.

CASUAL RIDERS ARE 10X MORE LIKELY THAN MEMBERS TO RETURN TO THEIR STARTING STATION

Bar Chart of the share of rides that are round trips.



- Round trip = a ride that starts and ends at the same station.

- 18.8% of casual rides start and end at the same station, compared with 1.9% of member rides.
- Most rides in both groups are point-to-point (about 81% for casual riders and 98% for members).

- Round trips are consistent with leisure or sighseeing rides, while point-to-point trips are consistent with commuting.

RECOMMENDATIONS: CONVERT CASUAL RIDERS INTO ANNUAL MEMBERS

- Run membership promotions at the stations casal riders use most. Use signage, QRQ codes or in app offers at those locations.

  Casual and Member riders start at largely different stations. Only one station appears in both top 10 lists, and casual riders' top stations include visitor locations such as Shedd Aquarium.

- Time digital campaigns for weekends and afternoons. Schedule social meadia and email pushes around the times casual riders are active.

  Casual riders make up a larger share on weekend rides and peak in the afternoon, while members peak at commute hours.

- Pitch membership around leisure value, and consider testing a weekend or flexible membership option. Show casual riders the savings if they ride regularly.

  Casual rides are about 3x longer (22 vs 8 minutes median), and 18.8% are round trips versus 1.9% for members, which is consistent with recreational use.

- Next steps: Validate these findings on a full 12 months of data to check seasonality. Add data on pricing, trip purpose or rider surveys to learn why casual riders choose single passes, and pilot the recomendations with a small campaign before a full rollout.

Hilda Yabasa ● October 29th 2026

THANK YOU